Technical Information

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Kollicoat® IR Coating Systems

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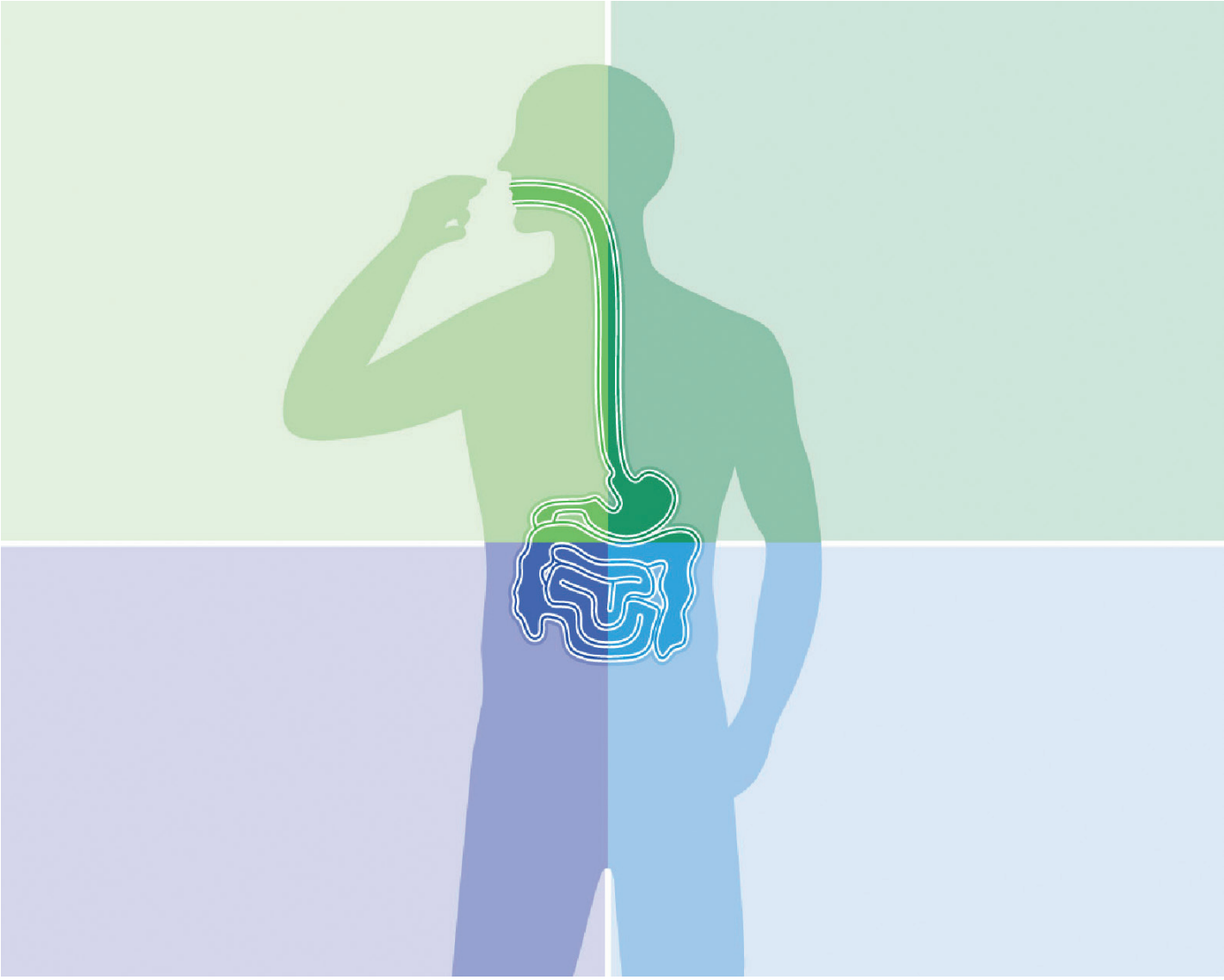
Ready-to-Use Coating Systems for Instant-Release Dosage Forms



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Contents

	Page
1. Introduction	3
1.1 General	3
1.2 Structural formula	4
1.3 Composition	4
1.4 Physical form	4
2. Properties and specifications	5
2.1 Chemical nature	5
2.2 Physicochemical properties	5
2.3 Properties of aqueous suspensions	7
2.4 Film properties	9
3. Application and processing	10
3.1 Applications	10
3.2 Incompatibilities	10
3.3 Processing notes	10
4. Process and formulation examples	11
4.1 Manesty XL Lab	11
4.2 Accela Cota	12
4.3 Thomas Engineering Accela-Cota	13
4.4 BGB-5C High Efficient Coating Machine	14
4.5 Up scaling characteristics in solid wall and side vented pan coaters	15
5. Storage conditions	16
6. Stability	16
7. Toxicology	16
8. Article-Nos.	16
9. Packaging	16
10. Note	16

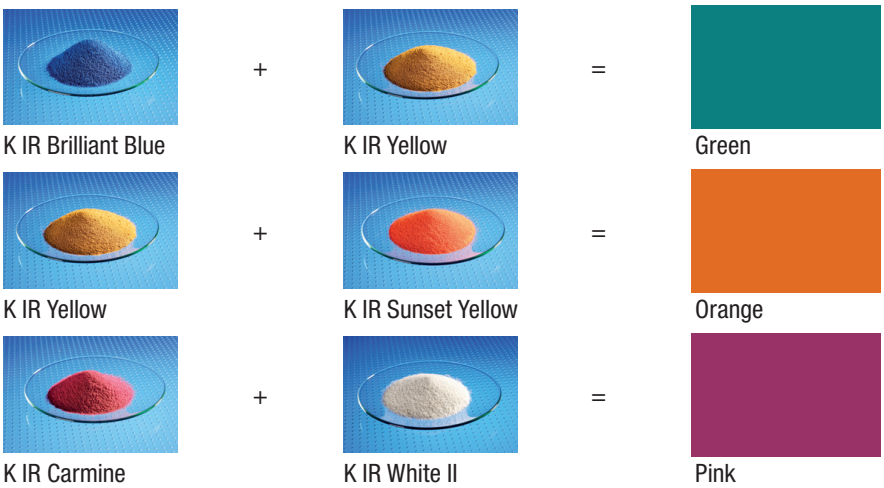
1. Introduction

1.1 General

Kollicoat® IR Coating Systems are the modular system for manufacturing of instant-release color coated tablets,
- using globally registered colors,
- shortening processing time,
- and minimizing logistic effort.

Kollicoat® IR Coating Systems are ready-to-use colored coating materials based on Kollicoat® IR (polyvinyl alcohol polyethylene glycol graft copolymer). This unique product line consists of five individual colored components, completed by a black and a white coating material:
- Kollicoat® IR Brilliant Blue
- Kollicoat® IR Carmine
- Kollicoat® IR Sunset Yellow
- Kollicoat® IR Red
- Kollicoat® IR Yellow
- Kollicoat® IR Black
- Kollicoat® IR White II

From the seven base components, it is possible to obtain a wide variety of color shades simply by combining different base colors in different concentrations. Few examples can be seen below.

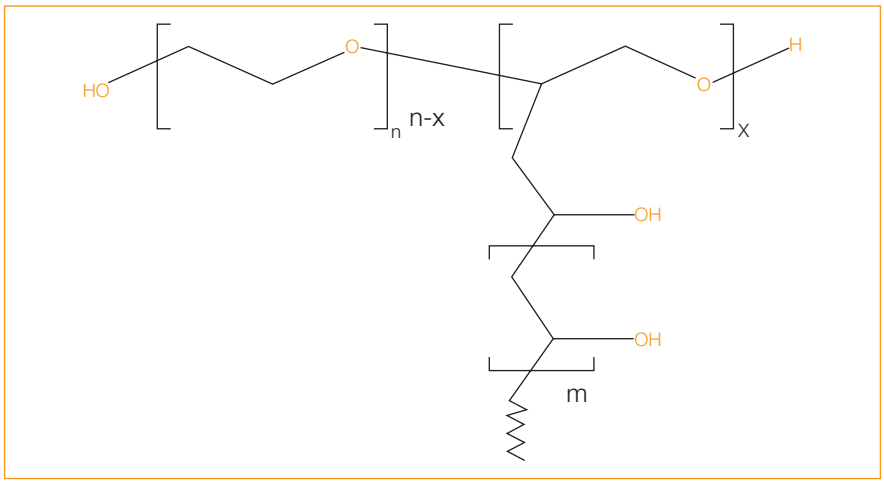


Due to its unique manufacturing process Xfast®, Kollicoat® IR Coating Systems can be redispersed rapidly in water.

The coating suspension is easily prepared by weighing and dispersing the required individual components in water. Without further additives the suspension is ready for being applied onto tablets.

1.2 Structural formula

Kollicoat® IR Coating Systems are based on a graft copolymer of PVA & PEG with the following structure:



PEG is the backbone of the structure and acts as an internal plasticizer.

1.3 Composition

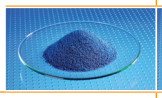
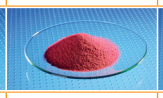
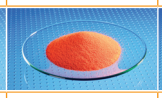
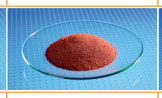
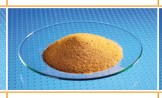
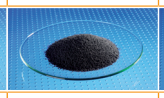
	Kollicoat® IR						
	Brilliant Blue	Carmine	Sunset Yellow	Red	Yellow	Black	White II
							
Kollicoat® IR	approx. 58.1%	approx. 58.1%	approx. 58.1%	approx. 58.1%	approx. 58.1%	approx. 58.1%	approx. 58.1%
Kollidon® VA 64	approx. 6.4%	approx. 6.4%	approx. 6.4%	approx. 6.4%	approx. 6.4%	approx. 6.4%	approx. 6.4%
Titanium dioxide	approx. 8.4%	approx. 8.4%	approx. 8.4%	approx. 8.4%	approx. 8.4%	approx. 8.4%	approx. 18.3%
Kaolin	approx. 15.3%	approx. 15.3%	approx. 15.3%	approx. 15.3%	approx. 15.3%	approx. 15.3%	approx. 15.3%
Sodium lauryl sulfate	approx. 1.9%	approx. 1.9%	approx. 1.9%	approx. 1.9%	approx. 1.9%	approx. 1.9%	approx. 1.9%
Brilliant Blue	approx. 9.9%	–	–	–	–	–	–
Carmine	–	approx. 9.9%	–	–	–	–	–
Sunset Yellow	–	–	approx. 9.9%	–	–	–	–
Iron oxide red	–	–	–	approx. 9.9%	–	–	–
Iron oxide yellow	–	–	–	–	approx. 9.9%	–	–
Iron oxide black	–	–	–	–	–	approx. 9.9%	–

Table 1: Composition of Kollicoat® IR Coating Systems

1.4 Physical form

Kollicoat® IR Brilliant Blue are light blue to blue free-flowing granules.
Kollicoat® IR Sunset Yellow are orange free-flowing granules.
Kollicoat® IR Carmine are light purple to purple free-flowing granules.
Kollicoat® IR Red are brownish-red free-flowing granules.
Kollicoat® IR White II are white to off-white free-flowing granules.
Kollicoat® IR Yellow are yellow free-flowing granules.
Kollicoat® IR Black are gray to black free-flowing granules.

2. Properties and specifications

2.1 Chemical nature

Owing to the special manufacturing process for Kollicoat® IR Coating Systems, the fine pigments are embedded in a polymer matrix and therefore unable to separate or agglomerate.

Three products are based on aluminum lakes: Kollicoat® IR Brilliant Blue, Kollicoat® IR Carmine, and Kollicoat® IR Sunset Yellow. Three products are based on iron oxides: Kollicoat® IR Red, Kollicoat® IR Yellow, and Kollicoat® IR Black. Kollicoat® IR White II, which contains additional titanium dioxide, completes the product line.

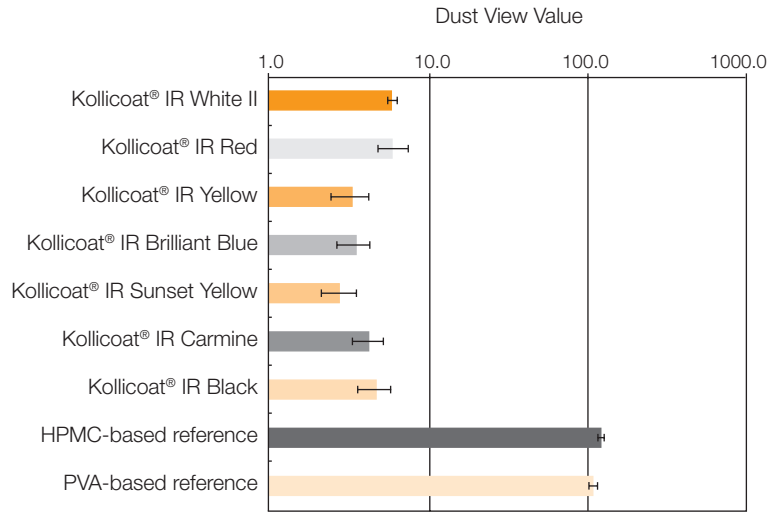
2.2 Physicochemical properties

Kollicoat® IR coating system are prepared by the Xfast® technology (spray agglomeration). This process generates granules with robust raspberry structure with high surface area & the following characteristics:
1. Uniform particle size distribution
2. No dust formation
3. Free flowing
4. No lumping
5. No segregation

Particle size distribution Kollicoat® IR coating systems granules show a narrow & consistent particle size distribution.

Dust formation

The nearly complete absence of dust makes the handling of Kollicoat® IR Coating Systems very easy and safe.



Dust content [%]	< 0.01	0.01 – 0.05	0.05 – 0.1	> 0.1
Dust View Value	< 1	1 – 12	12 – 25	> 25
Evaluation	Dust-free	Nearly dust-free	Essentially non-dusty	Very dusty

Figure 1: Dust View Values for Kollicoat® IR Coating Systems and references.

Flowability

Kollicoat® IR Coating system has very good flowability with the angle of repose around 30 °C, imparted by the granular system.

Redispersibility

During the Xfast® manufacturing process for Kollicoat® IR Coating Systems, the fine pigments are embedded in the Kollicoat® IR polymer matrix – a solid dispersion is formed. Therefore the pigments are unable to separate or agglomerate. The raspberry-like structure due to the innovative drying process and the water-soluble matrix polymer enable extremely rapid redispersibility in water.

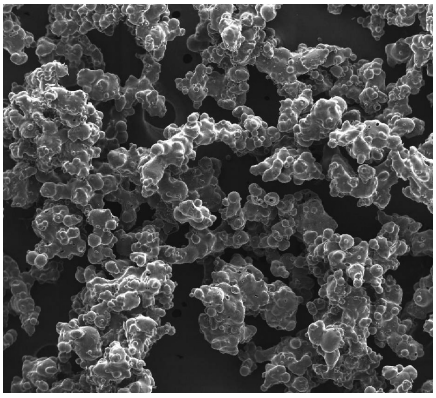


Figure 2: SEM Kollicoat® IR Red. *SEM from powder blend*

Within approximately 15 minutes the suspension is prepared and ready for starting the coating process. For the preparation of the suspension low shear mixing is recommended.

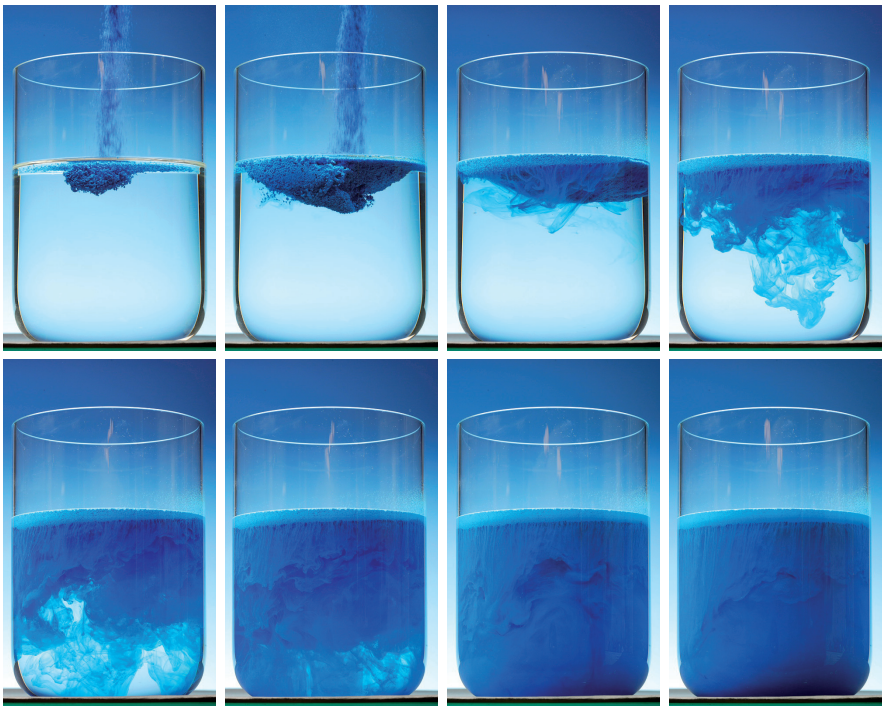


Figure 3: Dispersion of Kollicoat® IR Brilliant Blue in water.

Specification

See separate document: “Standard Specification (not for regulatory purposes)” available via BASF’s WorldAccount: https://worldaccount.basf.com (registered access).

*Kollicoat® IR Coating System	Colorant
Kollicoat® IR White II	Titanium dioxide
Kollicoat® IR Black	Ferric oxide, black
Kollicoat® IR Red	Ferric oxide, red
Kollicoat® IR Yellow	Ferric oxide, yellow
Kollicoat® IR Brilliant Blue	FD&C Blue No. 1
Kollicoat® IR Sunset Yellow	FD&C Yellow No. 6
Kollicoat® IR Carmine	Carmine

Regulatory status

At BASF we are committed to making regulatory affairs easier for you: A US Drug Master File "Kollicoat® IR Coating Systems (# 23187, Type IV) is already assigned by the FDA."

Kollicoat® IR (film-forming polymer in Kollicoat® IR Coating Systems) was approved as coating agent in medicinal products in the U.S., the EU, and in Japan. The polymer is also approved and used as coating agent in drug products in several other countries all over the world.

The pigments in Kollicoat® IR Coating Systems are globally registered.

Pharmacopoeial monographs

Kollicoat® IR	USP, Ph.Eur., JPE
Kollidon® VA 64	USP, Ph.Eur., JPE
Titanium dioxide	USP, Ph.Eur., JP
Kaolin	USP, Ph.Eur., JP
Carmine	JPE
Iron oxides	USP (Ferric oxide), JPE
Sodium lauryl sulfate	USP, Ph.Eur., JP

2.3 Properties of aqueous suspensions

Viscosity

The viscosity of aqueous dispersions of Kollicoat® IR Coating Systems increases with the polymer concentration, but remains very low compared to other polymers. Polymer suspensions with viscosities below 200 mPa·s ensure smooth coating processes. With Kollicoat® IR Coating Systems such low viscosities are achieved even at high solid contents of 25 – 30% in the coating suspension.

Systems based on PVA- and HPMC are typically recommended to be handled below 20% and 15%, respectively.

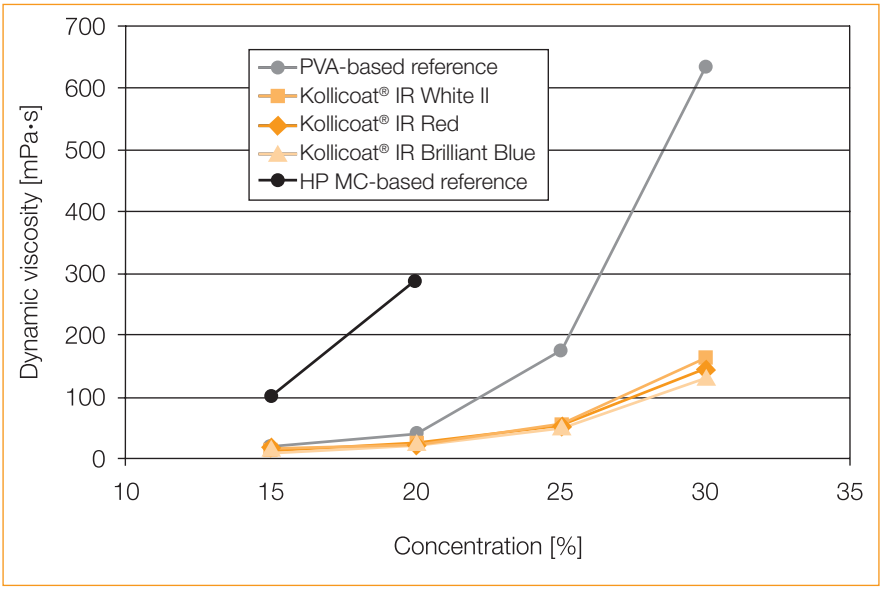


Figure 4: Dynamic viscosities of Kollicoat® IR Coating Systems and references.
Figures 5 and 6 show viscosities of Kollicoat® IR Red and Kollicoat® IR Brilliant

Blue suspensions as a function of solids concentration. Due to viscosities below 200 mPa·s at 30% solids contents, smooth coating processes are still enabled. Viscosities only depend on solids content but not on the kind of base color as proved in figures 5 and 6.

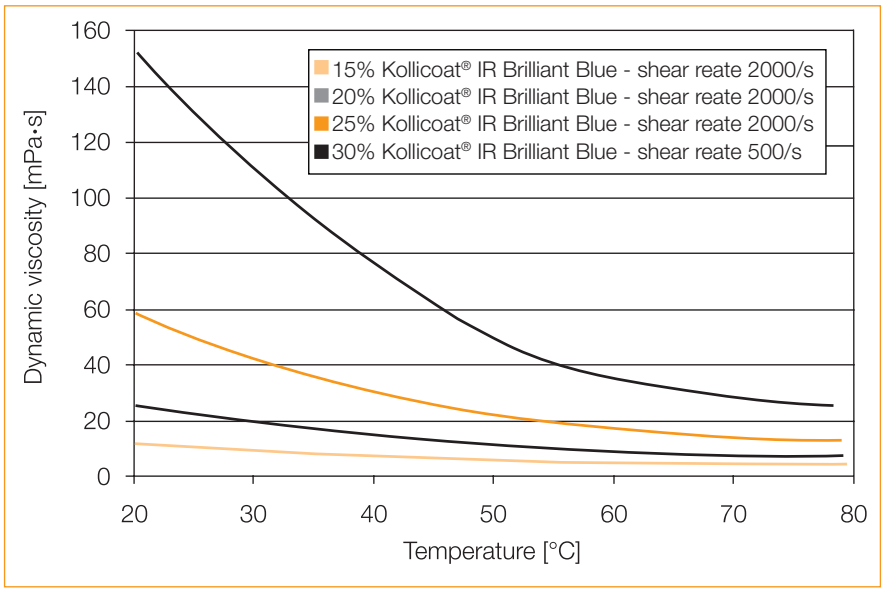


Figure 5: Viscosities of Kollicoat® IR Blue suspensions as a function of solids content.

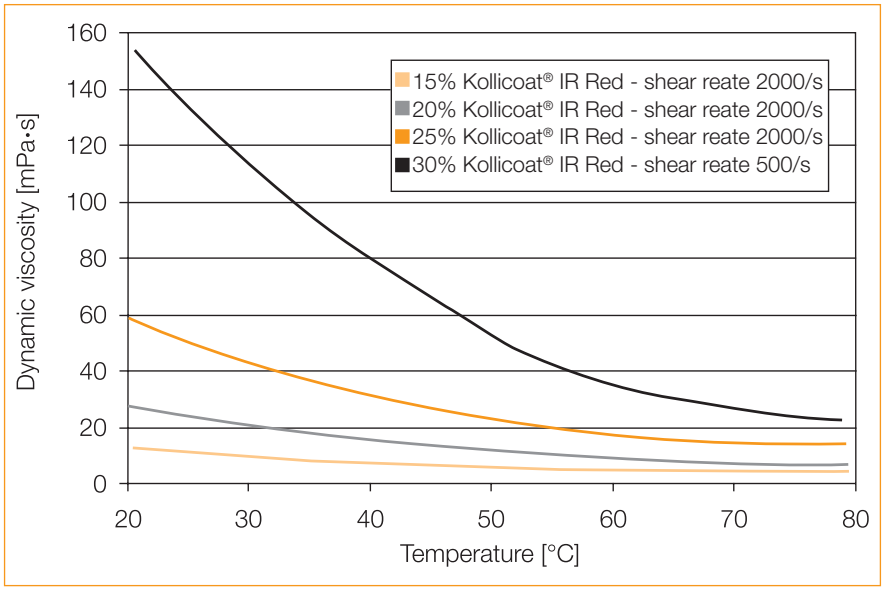


Figure 6: Viscosities of Kollicoat® IR Red suspensions as a function of solids content.

Surface tension

The surface tension of a 20% Kollicoat® IR Coating Systems suspension is very low – around 40 mN/m. Good wetting and spreadability are therefore easily achievable even on lipophilic surfaces.

2.4 Film properties

Kollicoat® IR Coating Systems form smooth, highly flexible colored films that disperse very rapidly in water. These films are not tacky and can readily be printed.

Plasticizer

Kollicoat® IR Coating Systems are ready-to-use colored coating materials based on Kollicoat® IR. Kollicoat® IR is a polyvinyl alcohol polyethylene glycol graft copolymer. Its backbone of polyethylene glycol acts as plasticizer (1.2). Due to the covalent bonding of the softening agent no migration of the plasticizer will occur; the tablets will not get tacky during their storage. The addition of further plasticizer is not necessary.

Elongation at break

Kollicoat® IR leads to flexible films. They are significantly more flexible than those made of a comparable polyvinyl alcohol-based product. The high flexibility of the films ensures that they do not crack on the tablets.

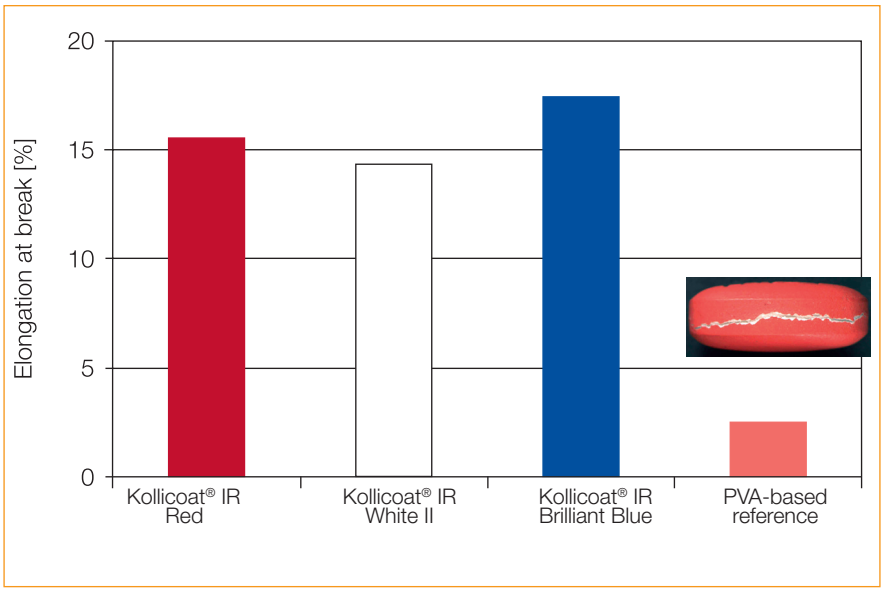


Figure 7: Elongation at break of various ready-to-use instant-release films (23 °C, 52% r. h.).

Coating engravings

The very low viscosity as well as excellent wetting and adhesion to the tablet surface ensure that even fine engravings are uniformly coated. No logo bridging occurs. The scanning electron micrograph below shows a tablet coated with Kollicoat® IR White II at a coating level of 5 mg/cm^2.

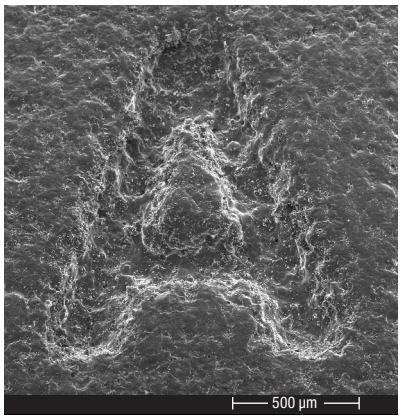


Figure 8: SEM of coating engraving.

3. Application and processing

3.1 Applications

Kollicoat® IR Coating Systems can be used in all applications where a readily redispersible, flexible, colored instant-release tablet coating is required.

Kollicoat® IR-based coating systems can be applied without plasticizer and with high solids contents up to 30%. This is the reason why they help to significantly reduce the process time and contribute to an efficient and economic manufacturing process.

3.2 Incompatibilities

Ibuprofen may lead to discoloration of pure Kollicoat® IR White II coatings when film-coated tablets are stored under light. This yellowing effect is caused by an incompatibility of ibuprofen with the Kaolin contained in Kollicoat® IR White II.

A yellowing will not be observable when colored coatings are used. Therefore, the use of colored components from Kollicoat® IR Coating Systems is recommended.

3.3 Processing notes

1. Because of the high flexibility of Kollicoat® IR Coating Systems films, it is not necessary to add a plasticizer (see 2.4 Film properties).
2. High shear during the preparation of coating suspension should be avoided in order to prevent formation of air bubbles.
3. It is recommended to use KCCS at solids concentrations of 25% to 30%.
4. No magnetic stirrer should be used when Kollicoat® IR Black is present in the coating suspension.

Preparation of suspension

A spray suspension is conveniently prepared as follows:
Weigh in the required amount of Kollicoat® IR Coating Systems components needed for your desired shade of color. Stir all components into water at low shear in order to disperse. The mixer speed should be adjusted so that little or no foam is produced. After stirring for approximately 15 min under low shear, the suspension is ready for further processing (see 2.2 Physicochemical properties).

Due to the low viscosity of aqueous dispersions, Kollicoat® IR Coating Systems can be sprayed at much higher concentrations than other ready-to-use instant-release formulations (see 2.3 Properties of aqueous suspensions). Suspensions with solids concentrations 25 – 30% can be processed without difficulty using different types of coating equipment.

Kollicoat® IR Coating Systems can be applied on all the common coaters, e.g. horizontal drum coaters, fluidized bed coaters, immersion sword coaters, and coating pans, under the usual conditions for aqueous suspensions.

The following conditions have produced good results in numerous trials:

Inlet air temperature	50 – 80 °C
Outlet air temperature	30 – 50 °C
Atomizing pressure	2 – 4 bar
Temperature of spray suspension	20 – 70 °C

Cleaning

All dry parts should be vacuum-cleaned first to remove all dry colored particles. Then the remaining product can very easily be rinsed off the equipment with cold or warm water.

4. Process and formulation examples

4.1 Manesty XL Lab

Formulation No. 01

Composition of placebo cores

79.25% Ludipress® LCE, 20% Kollidon® VA 64, 0.75% magnesium stearate

Composition of spray suspension

The formulation is designed for 5 kg of tablets (tablet weight 350 mg, diameter 9 mm).

	Weight [g]	Proportion [%]
Spray suspension		
Kollicoat® IR White II	50	5
Kollicoat® IR Carmine	150	15
Water	800	80
	1,000	100

Preparation of spray suspension

Spray suspension:
Stir the Kollicoat® IR Coating Systems into the specified quantity of water.
After 15 min, the spray suspension is ready for further processing.

Machine parameters

Coating machine	Manesty XL Lab 01 drum coater (480 mm)
Batch size	5 kg
Pan size	480 mm
Inlet air temperature	55 °C
Outlet air temperature	41 °C
Product temperature	36 °C
Outlet air flow	480 m^3/h
Atomizing pressure	2.4 bar
Forming air pressure	2.4 bar
Number of spray nozzles	1 (diameter 0.8 mm)
Spraying rate	31 g/min
Spraying time	40 min
Final drying	60 °C/4 min
Quantity applied	5.9 mg/cm^2 solids

Tablet properties

	Core	Film-coated tablet
Appearance	White	Purple
Hardness	91 N	75 N
Friability	0%	0%
Disintegration time	3:17 [min:s]	3:29 [min:s]

4.2 Accela Cota

Formulation No. 02

Composition of placebo cores

79.25% Ludipress LCE, 20% Kollidon VA 64, 0.75% magnesium stearate

Composition of spray suspension

The formulation is designed for 5 kg of tablets (tablet weight 350 mg, diameter 9 mm).

	Weight [g]	Proportion [%]
Spray suspension		
Kollicoat® IR Yellow	125	12.5
Kollicoat® IR Brilliant Blue	125	12.5
Water	750	75
	1,000	100

Preparation of spray suspension

Spray suspension:
Stir the Kollicoat® IR Coating Systems into the specified quantity of water. After 15 min, the spray suspension is ready for further processing.

Machine parameters

Coating machine	Accela drum coater (24-inch)
Batch size	5 kg
Pan size	24 inches
Inlet air temperature	55 °C
Outlet air temperature	42 °C
Product temperature	38 °C
Outlet air flow	414 m^3/h
Atomizing pressure	2.4 bar
Forming air pressure	1.4 bar
Number of spray nozzles	1 (diameter 0.8 mm)
Spraying rate	27 g/min
Spraying time	35 min
Final drying	60 °C/4 min
Quantity applied	5.9 mg/cm^2 solids

Tablet properties

	Core	Film-coated tablet
Appearance	White	Green
Hardness	87 N	79 N
Friability	0%	0%
Disintegration time	3:04 [min:s]	3:17 [min:s]

4.3 Thomas Engineering Accela-Cota

Formulation No. 03

Composition of placebo cores 79.25% Ludipress LCE, 20% Kollidon VA 64, 0.75% magnesium stearate

Composition of spray suspension The formulation is designed for 1 kg of oblong tablets (tablet weight 450 mg, 7 mm x 15 mm).

	Weight [g]	Proportion [%]
Spray suspension		
Kollicoat® IR Red	10	5
Kollicoat® IR White II	30	15
Water	160	80
	200	100

Preparation of spray suspension

Spray suspension:
Stir the Kollicoat® IR Coating Systems into the specified quantity of water. After 15 min of low shear mixing, the spray suspension is ready for further processing.

Machine parameters

Coating machine	15" Thomas Engineering Accela-Cota
Batch size	1 kg
Pan size	15 inches
Inlet air temperature	26 – 55 °C
Outlet air temperature	25 – 52 °C
Product temperature	22 – 50 °C
Outlet air flow	150 – 165 CFM
Atomizing pressure	2.4 bar
Forming air pressure	2.4 bar
Number of spray nozzles	1
Spraying rate	2 – 13
Spraying time	15 – 100 min
Final drying	5 min
Quantity applied	200 g

Tablet properties

	Core	Film-coated tablet
Appearance	White, oblong	Red, oblong
Hardness	147 N	162 N
Friability	0.17%	0.03%
Disintegration time	16.5 min	15.3 min

4.4 BGB-5C High Efficient Coating Machine

Formulation No. 04

Composition of placebo cores

TCM tablets

Composition of spray suspension

The formulation is designed for 4 kg of tablets (tablet weight 508 mg, diameter 12 mm).

	Weight [g]	Proportion [%]
Spray suspension		
Kollicoat® IR Yellow	28.8	4
Kollicoat® IR White II	115.2	16
Water	576	80
	720	100

Preparation of spray suspension

Spray suspension:
Stir the Kollicoat® IR Coating Systems into the specified quantity of water.
After 15 min, the spray suspension is ready for further processing.

Machine parameters

Coating machine	BGB-5C
Batch size	4 kg
Pan size	5 l
Inlet air temperature	50 – 55 °C
Outlet air temperature	38 – 42 °C
Product temperature	35 – 38 °C
Outlet air flow	350 m^3/h
Atomizing pressure	2.6 bar
Forming air pressure	2.6 bar
Number of spray nozzles	1
Spraying rate	20 g/min
Spraying time	30 min
Final drying	10 min
Quantity applied	600 g

Tablet properties

	Core	Film-coated tablet
Appearance	Yellowish-brown round tablets with spots	Uniform yellow round tablet
Hardness	152 N	187 N
Friability	0.13%	0%
Disintegration time	28 min	30 min

4.5 Up scaling characteristics in solid wall and side vented pan coaters

Composition of cores

Two types of tablet cores differing in regard to composition, shape & size were used the trails.

a. Round shaped tablets:
Wt – 300 mg, surface – 212 mm^2 Ludipress® – 79.5%, Kollidon® VA 64 – 20.0%, Mg-stearate – 0.5%

b. Football shaped tablets :
Wt – 450 mg, surface – 310 mm^2 Caffeine – 15.5%, Ludipress® LCE – 74.0%, Kollidon® CL-F – 5.0%, Kollidon® VA 64 fine – 5.0%, Mg-stearate – 0.5%

Machine parameters

Side vented pan coater

Parameters	Optimal settings Perfima Lab	Calculated settings Perfima 200
Batch size	40 kg	140 kg
Number of nozles	2	4
Drum speed	7 rpm	5 rpm
Circumferential speed	361 mm/s	349 mm/s
Inlet air temperature	50 °C	50 °C
Inlet air quantity	1060 m^3/h	2120 m^3/h
Maximum spray rate	150 g/min	300 g/min

Solid wall coater

Parameters	Optimal settings GS HP 25	Calculated settings GS 300
Batch size	22 kg	202 kg
Number of nozles	1	4
Drum speed	8 rpm	4 rpm
Circumferential speed	335 mm/s	329 mm/s
Inlet air temperature	60 °C	60 °C
Inlet air quantity	350 m^3/h	1400 m^3/h
Maximum spray rate	48 g/min	192 g/min

Result

Keeping the circumferential speed of the drum, spray rate per nozzle & the ratio of process air volume per nozzle constant was an appropriate measure for scale up. This is of special importance for thermolabile & moisture sensitive formulations, as this kind of scale up assures that the moisture & temperature exposure of the cores are kept as constant as possible. Hence the stability data gathered for cores coated in small scale are valid for tablets coated in large scale as well.

5. Storage conditions

Store at room temperature.

6. Stability

Ongoing stability studies according to ICH guidelines document stabilities of at least 36 months at room temperature in the original sealed packaging at this point in time:

Kollicoat® IR Brilliant Blue	36 months
Kollicoat® IR Carmine	36 months
Kollicoat® IR Sunset Yellow	36 months
Kollicoat® IR Red	36 months
Kollicoat® IR Yellow	36 months
Kollicoat® IR Black	36 months
Kollicoat® IR White II	36 months

7. Toxicology

The safety of Kollicoat® IR as a film coating polymer for medicinal products is supported by all relevant non-clinical (toxicological) studies. A summary is available on request. A detailed report can be shared as part of a non-disclosure agreement.

8. Article-Nos.

Kollicoat® IR Coating Systems	PRD-Nos.	Article-Nos.		
		25 kg carton box	1.5 kg canister[4]	0.3 kg bottle[4]
Kollicoat® IR Brilliant Blue	30371256	51683363	57376463	53750098
Kollicoat® IR Carmine	30371263	51683522	57377046	53750416
Kollicoat® IR Sunset Yellow	30371261	51683469	57376675	53750628
Kollicoat® IR Red	30371193	51683734	57376834	53750575
Kollicoat® IR Yellow	30483932	51684317	57376940	53749674
Kollicoat® IR Black	30488712	51684370	57376728	53749992
Kollicoat® IR White II	30371683	51683893	57376622	53750734

[4] Sample for technical use only

9. Packaging

103 l carton box with aluminum-laminated inliner, 25 kg capacity
5 l canister, 1.5 kg capacity (sample for technical use only)
1 l bottle, 0.3 kg capacity (sample for technical use only)

10. Note

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